

MOSKOVA DOMODEDOVO, Russian Federation

WMO: 370008

Lat: 55.409N

Lon: 37.906E

Elev: 588

StdP: 14.39

Time Zone: 3.00 (E03)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-5.8	-16.7	2.3	-11.1	-11.0	3.1	-5.7	22.1	27.1	20.3	26.7	4.4	40	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	86.3	67.1	82.7	65.8	79.2	64.6	69.8	80.5	68.1	78.7	66.7	76.6	8.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	99.3	73.8	64.6	93.2	71.9	62.9	87.7	70.5	34.1	80.6	32.8	78.8	31.6	76.0	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	18.3	16.4														
			DB		-16.8	89.6	6.8	3.8	-21.7	92.3	-25.6	94.5	-29.4	96.7	-34.4	99.4
			WB		-17.2	72.7	6.7	2.2	-22.0	74.3	-25.9	75.6	-29.7	76.9	-34.5	78.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.6	18.1	19.3	28.9	43.9	55.4	61.4	67.2	63.7	53.5	42.3	31.5	24.2	
	DBStd	19.23	12.54	12.48	9.10	8.44	8.46	6.91	6.01	6.48	6.59	7.90	9.03	10.71	
	HDD50	4423	988	860	654	222	44	4	0	0	37	259	555	800	
	HDD65	8411	1453	1280	1119	634	312	148	45	102	346	702	1005	1265	
	CDD50	1720	0	0	0	37	212	347	534	425	143	22	0	0	
	CDD65	234	0	0	0	0	14	41	114	62	3	0	0	0	
	CDH74	2445	0	0	0	7	204	475	1091	632	36	0	0	0	
	CDH80	709	0	0	0	1	37	118	348	203	2	0	0	0	
Wind	WSAvg	8.2	9.0	8.8	9.0	8.0	7.9	7.7	7.1	7.2	7.3	8.4	8.8	9.4	
Precipitation	PrecAvg	26.2	1.9	1.5	1.4	1.3	2.2	2.8	3.1	2.9	2.6	2.6	2.0	1.8	
	PrecMax	34.6	3.3	3.1	2.9	2.2	4.6	4.6	6.6	5.7	7.3	5.3	3.4	3.1	
	PrecMin	17.2	0.6	0.3	0.6	0.5	0.7	0.4	0.5	0.5	0.5	0.6	0.1	0.6	
	PrecStd	4.3	0.7	0.6	0.6	0.5	1.0	1.0	1.6	1.5	1.3	1.4	0.7	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.8	39.3	57.0	74.8	84.3	87.6	93.1	93.3	78.6	67.8	53.8	46.4	
		MCWB	39.0	37.2	46.4	55.5	65.2	66.4	68.3	67.2	61.8	56.5	50.8	44.9	
	2%	DB	37.3	37.2	48.4	68.2	79.1	83.9	87.6	84.6	73.6	61.1	48.4	42.5	
		MCWB	35.7	34.3	40.1	53.1	62.5	65.4	67.0	66.9	60.2	54.4	46.3	41.0	
	5%	DB	35.4	35.6	44.3	64.0	75.3	79.9	84.0	80.6	69.8	57.1	46.0	39.0	
		MCWB	34.2	33.7	37.9	50.9	59.8	64.1	66.6	65.4	59.7	52.2	44.2	37.6	
	10%	DB	33.7	34.0	39.6	59.2	71.6	75.4	80.4	76.8	65.8	53.6	42.6	35.8	
		MCWB	32.5	32.8	35.6	48.2	57.4	62.2	65.9	64.2	57.2	50.0	41.1	34.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	37.4	46.4	57.4	66.8	70.9	73.1	71.5	64.6	58.4	51.3	44.7
			MCDB	39.9	38.4	54.1	70.7	80.2	81.8	82.5	83.2	73.2	63.8	54.1	45.7
2%		WB	35.3	34.9	41.3	54.5	64.0	68.0	70.2	68.9	61.8	55.4	46.4	40.5	
		MCDB	36.2	35.7	46.3	64.6	75.7	78.3	80.4	79.6	69.9	60.0	47.7	41.4	
5%		WB	33.6	33.7	38.4	52.1	61.9	65.9	68.7	67.1	59.8	52.7	43.6	37.1	
		MCDB	34.4	34.5	42.5	61.6	71.6	75.6	79.2	77.5	67.5	56.9	44.8	38.1	
10%		WB	32.1	32.7	36.2	49.7	59.7	63.6	67.3	65.3	57.9	50.4	40.5	34.4	
		MCDB	33.0	33.8	39.3	57.6	68.3	72.5	77.2	74.7	64.5	53.3	41.7	35.1	
Mean Daily Temperature Range	5% DB	MDBR	9.7	11.6	13.8	18.5	21.0	20.0	19.8	19.2	16.6	11.2	7.8	7.9	
		MCDBR	6.7	8.3	18.3	25.8	26.5	26.4	25.8	25.3	23.6	17.7	11.1	7.2	
		MCWBR	6.7	7.3	11.7	13.7	12.1	11.7	9.5	10.3	12.2	11.4	9.4	7.1	
	5% WB	MCDBR	6.7	6.8	15.5	23.1	23.5	22.8	21.5	22.3	20.4	15.6	9.4	6.8	
		MCWBR	6.8	6.6	10.6	13.4	11.7	11.8	9.8	10.5	11.5	10.9	8.8	6.8	
Clear-Sky Solar Irradiance	taub		0.291	0.332	0.350	0.424	0.401	0.389	0.434	0.440	0.391	0.360	0.325	0.291	
	taud		2.111	2.049	2.102	2.101	2.290	2.345	2.242	2.207	2.325	2.340	2.232	2.122	
	Ebn at Noon		190	224	254	251	267	272	255	244	241	219	179	160	
	Edh at Noon		20	31	37	44	38	37	40	39	30	24	18	16	
All-Sky Solar Radiation	RadAvg		196	472	959	1273	1685	1744	1719	1401	888	410	160	107	
	RadStd		40	62	99	106	153	169	178	153	117	70	30	31	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.24	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+279	N/A	
(47)	Regional (7 neighbors)	+1.35	N/A	N/A	N/A	N/A	N/A	-283	-403	+209	N/A	

Nomenclature: See separate page